Polynomial Kernel

$a \begin{matrix} a_1 \\ a_2 \end{matrix}$ $b \begin{matrix} b_1 \\ b_2 \end{matrix}$

Remember : $[a^T b]^2 = [a_1\ a_2][b_1\ b_2] = (a_1 b_1 + a_2 b_2)^2 = a_1^2 b_1^2 + a_2^2 b_2^2 + 2a_1 b_1 a_2 b_2$

$|a^T b|^2$

$[a^T b + 1]^2$ $[a^T b + 1]^2 = (a^T b)^2 + 2a^T b + 1 = (a_1 b_1 + a_2 b_2)^2 = a_1^2 b_1^2 + a_2^2 b_2^2 + 2a_1 b_1 a_2 b_2 + 2(a_1 b_1 + a_2 b_2) + 1$

$= a_1^2 b_1^2 + a_2^2 b_2^2 + 2a_1 b_1 a_2 b_2 + 2a_1 b_1 + 2a_2 b_2 + 1$

$[a^T b + 1]^2 =$ $a_1^2\ \ a_2^2\ \ \sqrt{2}a_1 a_2\ \ \sqrt{2}a_1\ \ \sqrt{2}a_2\ \ 1$

$$\begin{matrix} b_1^2 \\ b_2^2 \\ \sqrt{2}b_1 b_2 \\ \sqrt{2}b_1 \\ \sqrt{2}b_2 \\ 1 \end{matrix} = \phi(a)^T \phi(b)$$

Where : $\phi(x) \equiv \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow$ $\begin{matrix} x_1^2 \\ x_2^2 \\ \sqrt{2}x_1 x_2 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ 1 \end{matrix}$

2D ⇒ 6D